MIL-F-21346B
5 August 1976
SUPERSEDING
MIL-F-21346A
16 July 1968
(See 6 8)

MILITARY SPECIFICATION

FUSEHOLDERS, BLOCK AND SHROUD TYPE

AND ASSOCIATED FUSE CLIPS,

GENERAL SPECIFICATION FOR

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1 SCOPE

1.1 Scope This specification covers block and shroud type fuseholders, and associated fuse clips for use with electrical, electronic, telephone, and instrument type fuses primarily covered by MIL-F-15160 and MIL-F-23419

1.2 Classification.

1 2.1 Type designation of fuseholders. The type designation of fuseholders shall be in the following form, and as specified (see 3.1 and 6 2)



1.2 1 1 Style The style is identified by the two-letter symbol "FH" followed by a two-digit number. The letters denote fuseholders, the number denotes the design and construction of a particular fuseholder

1 2 1 2 Class The class is identified by a two-letter symbol denoting the physical and electrical characteristics of a particular fuseholder. The first letter denotes the number of electrical poles in accordance with table I The second letter denotes the adaptability of the fuseholder to electrical circuits in accordance with table II The third letter "F", where applicable, shall designate fuseholders having copper clad steel fuse clips

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to Commander, Naval Ship Engineering Center, Center Building, SEC 6124, Prince George's Center, Hyattsville, Maryland 20782 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter

FSC 5920

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TABLE I Number of poles.

Symbol	Number of poles	Symbol	Number of poles
A	1	H	6
B	2	J	7
C	3	K	8
D	3 poles (2 poles fusible solid neutral bus)	L	9
E	4	M	10
F	4 poles (3 poles fusible, solid neutral bus)	N	11
G	5	P	12

TABLE II. Electrical circuits

Symbol	Electrical circuit
M	Main line
S	Single branch
D	Double branch

1 2 2 Type designation of fuse clips The type designation for fuse clips shall be in the following form and as specified (see 3.1 and 6.2)

FC21 A N

Style (1 2.2 1) Serial (1.2 2 2) Material (1.2 2 3)

1 2 2 1 Style The style shall be identified by the two-letter symbol "FC" followed by a two-digit number The letters denote fuse clips, the number denotes the design and construction of a particular fuse clip

1.2.2.2 Serial The serial for fuse clips within a particular style shall be designated by a single letter to identify different sizes

1.2 2.3 Material. Material for fuse clips is designated by a single letter "N" for nonferrous and "F" for copper clad steel

2 APPLICABLE DOCUMENTS

2 1 Issues of documents The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

HH-I-538 - Insulation, Electrical, Pasted-Mica
QQ-B-613 - Brass, Leaded and Non-Leaded Flat Products (Plate, Bar, Sheet, and Strip)
QQ-B-626 - Brass, Leaded and Non-Leaded· Rod, Shapes, Forgings, and Flat Products with Finished Edges (Bar and Strip).
QQ-B-750 - Bronze Phosphor, Bar, Plate, Rod, Sheet, Strip, Flat Wire, and Structural and Special Shaped Sections
QQ-C-502 - Copper Rods and Shapes· and Flat Products with Finished Edges (Flat Wire, Strips and Bars).
QQ-C-533 - Copper-Beryllium Alloy Strip (Copper Alloy Numbers 170 and 172)
QQ-C-576 - Copper Flat Products with Slit, Slit and Edge-Rolled, Sheared, Sawed, or Machined Edges (Plate Bar, Sheet, and Strip)
QQ-N-290 - Nickel Plating (Electrodeposited)
QQ-S-365 - Silver Plating, Electrodeposited, General Requirements for

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FEDERAL - Continued
QQ-S-571 - Solder, Tin Alloy, Lead-Tin Alloy, and Lead Alloy
QQ-S-763 - Steel Bars, Shapes, and Forgings - Corrosion Resisting
QQ-S-766 - Steel, Plate, Sheet, and Strip - Corrosion Resisting
QQ-W-423 - Wire, Steel, Corrosion-Resisting
PPP-B-566 - Boxes, Folding, Paperboard
PPP-B-636 - Box, Shipping, Fiberboard
PPP-B-676 - Boxes, Setup.

MILITARY
MIL-I-10 - Insulating Materials, Electrical, Ceramic, Class L
MIL-M-14 - Molding Plastics and Molded Plastic Parts, Thermosetting
MIL-P-116 - Preservation-Packaging, Methods of.
MIL-P-997 - Plastic-Material, Laminated, Thermosetting, Electric-Insulation, Sheets, Glass Cloth, Silicone Resin
MIL-T-10727 - Tin Plating, Electrodeposited or Hot-Dipped for Ferrous and Nonferrous Metals.
MIL-F-14256 - Flux, Soldering, Liquid (Rosin Base)
MIL-P-15037 - Plastic-Sheet, Laminated, Thermosetting, Glass-Cloth, Melamine-Resin
MIL-F-15160 - Fuses, Instrument, Power, and Telephone, General Specification for.
MIL-F-23419 - Fuses, Instrument Type, General Specification For
MIL-C-45662 - Calibration System Requirements

(See Supplement 1 for list of associated specification sheets.)

STANDARDS

MILITARY
MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-202 - Test Methods for Electronics and Electrical Component Parts
MIL-STD-831 - Test Reports, Preparation of.
MIL-STD-1285 - Marking of Electrical and Electronic Parts

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal, shall apply

NATIONAL BUREAU OF STANDARDS
Handbook H28 - Screw-Thread Standards for Federal Services.

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, D.C. 20402.)

NATIONAL MOTOR FREIGHT TRAFFIC ASSOCIATION, INCORPORATED, AGENT
National Motor Freight Classification

(Application for copies should be addressed to the American Trucking Association, Inc., Tariff Order Section, 1616 P Street, N.W., Washington, D.C. 20036.)

UNIFORM CLASSIFICATION COMMITTEE, AGENT
Uniform Freight Classification Rules

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 202 South Riverside Plaza, Chicago, IL, 60606.)

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3 REQUIREMENTS

3 1 Specification sheets The individual item requirements shall be as specified herein and in accordance with the applicable specification sheet. In the event of any conflict between requirements of this specification and the specification sheet the latter shall govern (see 6.2,

3 2 First article When specified (see 6 2), the contractor shall furnish sample units for first article inspection and approval (see 4.5 1 and 6 7)

3 3 Material The material for each part shall be as specified herein. However, when a definite material is not specified, a material shall be used which will enable the fuseholders and fuse clips to meet the performance requirements of this specification Acceptance or approval of any constitutent material shall not be construed as a guaranty of the acceptance of the finished product.

3 3.1 Restricted material. Flammable or explosive material, or material which can produce toxic or suffocating fumes when the fuseholders or fuse clips are in service shall not be used Mercury in any form shall not be used, including materials and parts thereof.

3 3.2 Plastic material. Plastic material shall be molded or laminated, as specified (see 3.1)

3.3 2.1 Molded Molded plastic shall conform to type MDG NME MSG. SDG-F or MFI-20 of MIL-M-14

3.3 2 2 Laminated Laminated plastic shall conform to MIL-P-997 or MIL-P-15037.

3 3 3 Metals. Metals shall be of a corrosion-resistant type or shall be treated to resist corrosion All metal parts shall be properly heat treated to enable them to meet the performance requirements specified herein (see 3.5).

3 3.3 1 Dissimilar metals (see 6 5) When dissimilar metals are used in intimate contact with each other, protection against electrolysis and corrosion shall be provided. The use of dissimilar metals in contact, which tend toward active electrolytic corrosion (particularly brass, copper, or steel used in contact with aluminum or aluminum alloy), is not acceptable However, plating of dissimilar base metals to provide similar abutting surfaces is permitted

3.3 3 2 Brass Brass shall conform to QQ-B-613 or QQ-B-626, as applicable.

3 3 3 3 Copper. Copper shall conform to QQ-C-502 or QQ-C-576, as applicable

3 3 3 4 Copper-beryllium alloy Copper-beryllium alloy shall conform to QQ-C-533.

3.3 3 5 Corrosion-resisting steel. Corrosion-resisting steel shall be hard or hardened, and shall conform to QQ-S-763 or QQ-S-766, as applicable

3.3 3 6 Corrosion-resisting steel wire Corrosion-resisting steel wire shall conform to QQ-W-423

3 3 3 7 Copper clad steel. Copper clad steel shall consist of a core, representing 80 percent, plus or minus 1 percent, of the sheet thickness, of SAE 1065, located between two layers of oxygen free copper. Each layer of the copper shall represent 10 percent, plus or minus 1 percent, of the total sheet thickness before plating.

3 3 3 8 Solder. Solder shall conform to QQ-S-571. Soldering fluxes, when used, shall conform to the requirements of MIL-F-14256.

3 3 3 9 Phosphor bronze. Phosphor bronze shall conform to QQ-B-750.

3 3 4 Mica. Mica shall conform to HH-I-538

3 3 5 Ceramic. Ceramic material shall conform to glazed brown, grade L211, of MIL-I-10

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3 3 6 Sealing material Sealing material shall be nonfungus nutrient or treated to resist fungus growth.

3 4 Design and construction Fuseholders and fuse clips shall be of the design, construction and physical dimensions specified (see 3 1)

3 4 1 Current-carrying parts Unless otherwise specified (see 3 1), all current-carrying parts shall be of nonferrous metal

3 4.2 Studs. Unless otherwise specified (see 6.2), all studs shall be brass.

3 4 3 Threaded parts Unless otherwise specified (see 3 1), all threaded parts shall be in accordance with handbook H28 Where practical, all threads shall be in conformance with the coarse-thread series The fine-thread series shall be used only for applications that might show a definite advantage through their use. Where a special diameter-pitch combination is required, the thread shall be of American National Form and of any pitch between 16 and 36 which is used in the fine-thread series

3.4.3 1 Engagement of threaded parts. All threaded parts shall engage by at least two full threads Where a screw mates with a plastic part, a threaded metal insert shall be molded therein.

3 4 4 Soldering. Solder shall not be used primarily to obtain mechanical strength. Connections shall be mechanically secure and electrically continuous before and after soldering. All excess flux shall be removed.

3 4 5 Plating (see 3 1) Unless otherwise specified, plating shall be not less than 0.0002-inch thick Finish shall be optional when not specifically designated

3.4 5 1 Silver plating Silver plating shall conform with grade B of QQ-S-365 (see 3 3 1)

3 4 5 2 Nickel plating Nickel plating shall be class 2, type VI (RC), of QQ-N-290

3 4 5 3 Bright alloy plating Bright alloy plating shall be of the following composition

	Percent
Copper	55 to 60
Tin	25 to 28
Zinc	14 to 18

3 4 5 4 Hot-tin-dip. The hot-tin-dip shall be in accordance with MIL-T-10727 (see 3.3 1)

3 4 6 Riveting Rivets shall be properly set and shall be flanged or rolled with no evidence of rivet fatigue. There shall be no evidence of cracking, chipping, peeling, denting or other deformation of the material riveted.

3 4 7 Terminal assemblies.

3.4 7 1 Lug terminals (solder type) All lug terminals (solder type) shall be hot-tin- or hot-solder-dipped to facilitate soldering

3 4 7 2 Terminal plates. Unless otherwise specified (see 3 1), all terminal plates for wire-binding screws shall be of metal not less than 0 047-inch thick If necessary, terminal-plate holes may be extruded to provide sufficient metal for engagement of two full threads (see 3 4 3.1)

3 4 7 3 Knife-blade terminal assemblies The design of knife-blade terminal assemblies shall be such as to permit the removal of attached wire lugs without loosening the fuse clip on the supporting or mounting base

3 4 8 Fuseholder assembly

3 4 8 1 Securing of assemblies Through-bolting shall be used when possible Lock-washers shall be incorporated under assembly-securing screws Flat washers shall be

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installed between lockwashers and plastic surfaces. Means other than friction between surfaces shall be employed to secure and rigidly fasten current-carrying parts to a supporting or mounting base. The mechanical means employed to secure parts to a supporting or mounting base, or to each other, shall be adequate to prevent loosening of the parts so secured when making wire connections.

3.4.8.2 Securing of knife-blade electrical clips. When knife-blade fuse clips are held in slots milled in the mounting piece, the slot-nested parts shall be sweated with solder and firmly secured with one or more rivets or screws.

3.4.8.3 Assembly of ceramic parts. Fungus-inert resilient material shall be placed against the bearing surfaces of ceramic parts to take up differential thermal expansion.

3.4.8.4 Sealing of securing screws, nuts, and rivets. All exposed live parts, such as screws, nuts, rivets, and the like, used to secure a fuse clip to a supporting or mounting base, shall be recessed not less than 0.09 inch. After assembly, the void shall be filled, covering the live parts to a depth of not less than 0.06 inch with sealing material. Sealing material shall not protrude past the surface of the base.

3.4.8.5 Mounting holes. Unless otherwise specified (see 3.1), mounting holes shall be so located and countersunk in multipole-fuseholder bases that there will be a space of not less than 0.50 inch over the surface of the insulating material between any mounting screw and the nearest live metal part. Where barriers or gutters are included in the design of the fuseholder, the 0.50-inch spacing requirement shall be the shortest insulating-surface path over the barrier or gutter.

3.4.8.6 Buses, straps, and terminals of main- and branch-circuit fuseholders. All buses and straps, including solid neutral straps, shall be copper. No uninsulated live metal terminals, bus bars, or other live metal parts shall be located within a wiring gutter. A solid neutral strap, with its line connection, shall be so covered or located that it shall not be exposed to accidental contact with adjacent circuit wires other than those connected thereto. Individual terminals shall be countersunk between close fitting walls or barriers to such a depth that when wired with an AWG size 10 solid conductor, the top of the screwhead shall not be in contact with a straightedge placed across the walls or barriers. More than one terminal in the same recess shall not be permitted, unless additional insulation is provided between such terminals.

3.5 Performance.

3.5.1 Insulation resistance. Resistance between mutually insulated parts shall be not less than 1,000 megohms (see 4.7.2).

3.5.2 Contact resistance. Total contact resistance of a pair of mounted fuse clips shall not exceed 0.005 ohm initially and not more than 0.010 ohm after endurance test (see 4.7.3).

3.5.3 Endurance. Fuseholders and mounted pairs of fuse clips shall remain electrically and mechanically operative for a minimum of 1,000 fuse insertions and withdrawals. The forces required after endurance cycling (see 4.7.4), shall be within 25 percent of the original forces. The contact resistance shall be within the limits specified in 3.5.2.

3.5.4 Thermal shock. There shall be no evidence of breaking, cracking, spalling, or loosening of terminals and inserts of the fuseholders plated copper clad steel parts shall not peel, flake or separate from base metal when subjected to the test of 4.7.5.

3.5.5 Vibration. Fuseholders and mounted pairs of fuse clips shall be capable of retaining an appropriate fuse while being tested in accordance with 4.7.6.

3.5.6 Shock. Fuseholders and mounted fuse clips shall retain an appropriate fuse when shock tested in accordance with 4.7.7 and there shall be no evidence of loose, broken or cracked parts.

3.5.7 Torque strength of inserts and threaded terminals. Inserts and threaded terminals shall not loosen, strip threads or deform, and fixed terminals shall not rotate when tested in accordance with 4.7.8.

3.5.8 Dielectric withstanding voltage. There shall be no breakdown, flashover, arcing or current flow in excess of 1 milliampere (mA) when subjected to the atmospheric pressure test of 4.7.9.1 and the reduced barometric pressure test of 4.7.9.2. The reduced barometric pressure test shall be performed only when specified (see 3.1).

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3 5 9 Moisture resistance Fuseholders having molded insulating material parts shall not show any evidence of breaking, cracking, spalling or loosening of terminals when tested as specified in 4.7.10 During the tenth cycle at a relative humidity of 90 to 98 percent and a temperature of 25 degrees Celsius (°C) ± 2°C, the insulation resistance shall be not less than 1 megohm. Following the 24-hour conditioning period, the insulation resistance shall be not less than 50 megohms.

3 5 10 Salt spray (corrosion) There shall be no flaking, peeling or loss of plated finish from the electrical contact surfaces of the fuse clips nor shall the base metal of noncurrent carrying parts be corroded after being subjected to salt spray test specified in 4 7 11

3 6 Marking. Unless otherwise specified (see 3 1), fuseholders and unmounted fuse clips shall be marked in accordance with MIL-STD-1285 with the following:

(a) Type designation (see 1.2 1, 1 2.3 and 3 1) 1/
(b) Manufacturer's name or code symbol.
(c) Additional marking as required (see 3 1)
(d) Fuse clips, when mounted integrally with fuseholders, need not be marked

1/Applicable to fuseholders only

3.7 Workmanship Fuseholders and fuse clips shall be uniform in quality, and shall be free from broken, cracked or displaced parts, burrs, sharp edges, spalling, peeling, flaking, and other defects which will affect life, serviceability or appearance

4. QUALITY ASSURANCE PROVISIONS

4 1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements

4.1.1 Test equipment and inspection facilities Test and measuring equipment and inspection facilities of sufficient accuracy, quality and quantity to permit performance of the required inspection shall be established and maintained by the contractor The establishment and maintenance of a calibration system to control the accuracy of the measuring and test equipment shall be in accordance with MIL-C-45662

4 1 1.1 Test fuse Test fuses required in some tests herein shall be as specified in the individual specification sheets for fuseholders and fuse clips (see 3 1) Ferrule and blade type fuses selected shall be to the minimum specified dimensions and alinement of the ferrules or blades shall be 50 percent of the specified tolerance (see 3 1)

4 1 1 2 Dummy fuse Dummy fuses required in some tests herein shall consist of bars, cylinders, or rods having the length and terminal diameter or blade dimensions required by MIL-F-15160 and MIL-F-23419 for corresponding specified fuses (see 3.1)

4 1 1.3 Submission of test reports and samples Unless otherwise specified (see 6 2), the tested items along with a report covering the inspection and tests for First article inspection and the Periodic Group C inspection of 4 6.2 shall be furnished to the procuring activity (see 6 2.1) The test reports shall be prepared in accordance with MIL-STD-831 The tested items shall be appropriately identified to indicate the test to which each was subjected

4 2 Classification of inspection The examination and testing of fuseholders and fuse clips shall be classified as follows (see 6 6)

(a) Materials inspection (see 4.3)
(b) First article inspection of fuseholders and associated fuse clips (see 4 5)
(c) Quality conformance inspection (see 4 6)

4 3 Materials inspection Materials inspection shall consist of certification (see 6 2.1) supported by verifying data that the materials listed in table III, used in fabricating the fuseholders and fuse clips, are in accordance with the applicable referenced specifications or requirements prior to such fabrication

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TABLE III Materials inspection

Material	Requirement paragraph	Applicable specification
Plastic	3 3 2	MIL-M-14 MIL-P-997 MIL-P-15037
Brass	3 3 3 2	QQ-B-613 QQ-B-626
Copper	3.3 3 3	QQ-C-502 QQ-C-576
Copper beryllium	3 3 3 4	QQ-C-533
Corrosion resisting steel	3 3 3.5	QQ-S-763, QQ-S-766
Corrosion resisting steel wire	3 3 3 6	QQ-W-423
Copper clad steel	3.3 3 7	----
Solder	3 3 3 8	QQ-S-571
Phosphor bronze	3 3 3 9	QQ-B-750
Mica	3 3.4	HH-I-538
Ceramic	3 3.5	MIL-I-10
Sealing material	3.3.6	----
Finishes		
Silver plate	3 4 5 1	QQ-S-365
Nickel plating	3.4.5 2	QQ-N-290
Bright alloy	3 4 5 3	----
Hot-tin-dip	3 4 5 4	MIL-T-10727

4 4 Inspection conditions Unless otherwise specified herein, all inspections shall be performed in accordance with the test conditions specified in the "General Requirements" of MIL-STD-202

4 5 First article inspection for fuseholders and associated clips First article inspection shall be performed in accordance with table IV

TABLE IV First article inspection

Examination or test	Requirement paragraph	Method paragraph
Group I (all sample units)		
Visual and mechanical examination	3 1, 3.3 to 3.4.8 6 incl, 3.6 and 3.7	4 7 1
Insulation resistance[1/]	3 5 1	4 7 2
Contact resistance	3 5 2	4 7 3
Group II (four sample units)		
Endurance	3 5 3	4 7 4
Thermal shock[1/]	3 5 4	4 7 5
Vibration	3 5 5	4 7 6
Shock	3 5 6	4 7 7
Torque strength[1/]	3 5 7	4 7 8
Dielectric withstanding voltage[1/]	3 5 8	4 7.9
Group III (four sample units)[1/] (Applicable to fuseholders having molded parts only)		
Moisture resistance	3 5 9	4 7 10
Dielectric withstanding voltage	3 5 8	4 7 9
Insulation resistance	3 5 1	4 7 2
Group IV (four sample units)		
Salt spray (corrosion)	3 5 10	4 7 11

[1/] Applicable to fuseholders only

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4 5 1 Sample size Samples for first article inspection shall consist of eight fuseholders or pairs of fuse clips for each part number, except 12 sample fuseholders shall be required for items having molded parts

4 5 2 Failures One or more failures shall be cause for refusal to grant first article approval

4 6 Quality conformance inspection.

4 6 1 Inspection of product for delivery. Inspection of product for delivery shall consist of groups A and B inspection.

4 6 1 1 Inspection lot An inspection lot shall consist of all fuseholders or fuse clips of the same part number produced under essentially the same conditions and offered for inspection at one time.

4.6.1.2 Group A inspection Group A inspection shall consist of the examination and tests specified in table V, and shall be made on the same set of sample units in the order shown

TABLE V. Group A inspection.

Examination or test	Requirement paragraph	Test paragraph	AQL		
				Major	Minor
Visual and mechanical examination	3.1, 3.3 to 3.5.8.6 incl, 3.6 and 3.7	4.7 1	-	1.0	2 5
Contact resistance	3 5.2	4.7.3	1.0	-	-
Dielectric withstanding voltage 1/	3 5.8	4.7 9	1.0	-	-

1/ Test applicable to fuseholders only

4 6 1 2 1 Sampling plan Sampling for group A inspection shall be in accordance with inspection level S-4 of MIL-STD-105 The acceptable quality level (AQL) shall be as specified in table V Major and minor defects shall be as specified in table VI Only two sample units shall be subject to verification of dimensional requirements.

TABLE VI. Classification of defects.

Examination	Major	Minor
Material not as specified	X	
Finish not as specified	X	
Fuse clip mounts not as specified	X	
Dimensions exceed tolerances	X	
Terminals loose, cracked, or displaced, insulating material broken	X	
Marking (type designation incorrect)	X	
Workmanship not as specified		X

4 6 1.3 Group B inspection Group B inspection shall consist of tests specified in table VII in the order shown and is applicable to fuseholders only

TABLE VII Group B inspection.

Test	Requirement paragraph	Test paragraph
Torque strength	3.5.7	4.7 8
Dielectric withstanding voltage	3.5 8	4 7 9

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4 6 1 3 1 Sampling plan Sampling for group B inspection shall be in accordance with MIL-STD-105, inspection level S-1 with an AQL of 4 percent defective The sample units shall be selected from each lot of those which have passed group A inspection

4 6 1 4 Disposition of sample units Sample items that have been subjected to and passed groups A and B inspection may be furnished on the contract

4 6 1 5 Rejected lots If an inspection lot is rejected, the contractor may withdraw the lot, rework it to correct the defects, or screen out the defective units, as applicable, and reinspect Such lots shall be kept separate from new lots and shall be clearly identified as reinspected lots Rejected lots shall be inspected using tightened inspection

4 6.2 Periodic inspection

4 6.2.1 Group C inspection. When required by the Command or Agency concerned, samples shall be selected and tested as specified in 4.5.1 and table IV for the Group C inspection The lot from which the samples were selected shall not be shipped until the samples have been approved by the cognizant activity and the lot is accepted by the government representative If however, any Group C test sample is rejected for noncompliance with the requirements of this specification, acceptance and shipment of any further fuseholders and fuse clips shall be withheld until the contractor can demonstrate that he has corrected the conditions which led to the rejection. Delivery of products which have passed the groups A and B inspection shall not be delayed pending the results of the group C inspection

4 6 2 1.1 Disposition of sample units Sample units which have been subjected to group C inspection shall not be furnished on the contract

4 7 Methods of examination and test.

4 7 1 Visual and mechanical (see 3 1, 3 3 to 3 4 8 6, 3 6 and 3 7) Fuseholders and fuse clips shall be examined to verify that the materials, design, construction, physical dimensions marking and workmanship are in accordance with the requirements specified herein

4 7.2 Insulation resistance (see 3 5 1) Unless otherwise specified (see 3.1), the insulation resistance of fuseholders shall be measured in accordance with method 302, test condition B, of MIL-STD-202. The following detail shall apply

Points of measurement - Between all terminals and ground and between all mutally insulated parts and unconnected terminals.

4 7.3 Contact resistance (see 3 5 2) Contact resistance of fuseholder and mounted fuse clips shall be measured in accordance with method 307 of MIL-STD-202. The following test conditions shall apply

(a) Fuse clips shall be tested in pairs or as assembled on fuseholders.
(b) Fuse clips shall be mounted by normal means with an appropriate dummy test fuse inserted
(c) Test voltage and current - 2 V, 1 ampere (A), d.c
(d) Resistance measurements - Shall be taken across fuseholder terminals or across the pairs of fuse clips. On fuseholders having three or more poles, only three poles need to be tested.

4 7 4 Endurance test (3 5.3). Test shall be conducted through the use of an appropriate dummy fuse One cycle shall be one complete insertion and complete withdrawal of the dummy fuse Fuse clips shall be tested in mounted pairs. Insertion and withdrawal forces and contact resistance shall be measured before and after test.

4 7 5 Thermal shock (see 3 5 4) Fuseholders shall be subjected to thermal shock in accordance with method 107 of MIL-STD-202. The following details and exceptions shall apply

(a) Test condition - A
(b) Measurements after cycling - The tested units shall be examined for loose, broken or cracked parts Plated and copper clad steel parts shall be examined for peeling, flaking or separation from the base materials.

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4.7.6 Vibration (see 3.5.5). Unless otherwise specified (see 3.1), fuseholders and mounted pairs of fuse clips shall be tested in accordance with method 204, test condition A, of MIL-STD-202. The following details shall apply.

(a) Method of mounting - Fuseholders or fuse clips shall be rigidly mounted by normal means.
(b) A test fuse of applicable size shall be inserted in the fuseholder or fuse clip.
(c) Fuses which open during test shall not be considered a failure of the fuseholder or fuse clip and shall be replaced and test continued to completion. Dislodging of test fuse during test shall be considered a failure.

4.7.7 Shock (see 3.5.6). Fuseholders or mounted pairs of fuse clips shall be tested in accordance with method 207 of MIL-STD-202. The following details and exceptions shall apply.

(a) Mounting fixture - Standard mounting fixture, figure 207-5.
(b) A test fuse of applicable size shall be inserted in the fusholder or fuse clip.
(c) Fuses which open during test shall not be considered a failure of the fuseholder or fuse clip and shall be replaced and test continued to completion. Dislodging of the test fuses during test shall be considered a failure.
(d) Measurement - Tested units shall be examined for loose, broken or cracked parts.

4.7.8 Torque strength of inserts and threaded terminals (see 3.5.7). Inserts and threaded terminals shall be tested in accordance with method 211, test condition E, of MIL-STD-202.

4.7.9 Dielectric withstanding voltage (see 3.5.8).

4.7.9.1 Atmospheric pressure (see 3.5.8). Fuseholders shall be tested in accordance with method 301 of MIL-STD-202. After test, the fuseholders shall be examined to determine conformance with 3.5.8. The following details and exceptions shall apply.

(a) Method of mounting - Normal mounting.
(b) Test voltage - Twice rated voltage plus 1,000 volts (V).
(c) Nature of potential - Alternating current.
(d) Points of application - Between each terminal and ground, between adjacent terminals of different poles, and between all unconnected terminals of the same pole.

4.7.9.2 Barometric pressure, reduced (see 3.5.8). Fuseholders shall be tested in accordance with method 105 of MIL-STD-202. Unless otherwise specified (see 3.1), the following details and exceptions shall apply.

(a) Test voltage - Twice rated voltage plus 500 V.
(b) Nature of potential - Alternating current.
(c) Test condition - C.

4.7.10 Moisture resistance (see 3.5.9). Fuseholders shall be subjected to method 106 of MIL-STD-202, including step 7a. The following details and exceptions shall apply.

(a) Mounting - Sample units shall be mounted vertically by normal means on a corrosion resistant metal panel.
(b) Polarization - During steps 1 to 6 inclusive, a d c potential of 100 V shall be applied between current carrying parts and mounting hardware. Negative polarity shall be applied to the mounting panel.
(c) Vibration, step 7b, shall be omitted.
(d) Final measurements - During the tenth cycle, insulation resistance measurements shall be made as specified in 4.7.2. Following a 24-hour period at 25° ± 2°C and 50 ± 5 percent RH, insulation resistance shall be measured as specified in 4.7.2.

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4 7 11 Salt spray (corrosion) (see 3 5 10) Fuseholders and fuse clips shall be subjected to salt spray test in accordance with method 101, MIL-STD-202 The following details and exceptions shall apply

(a) Test condition - A
(b) Measurement after test - The sample units shall be examined for flaking, peeling or other loss of finish plating on electrical contact surfaces Nonconducting ferrous parts shall be examined for disintegration of supporting parts.

4 7.12 Inspection of preparation for delivery Sample packages or packs and the inspection of the preservation, packaging and packing, and marking for shipment and storage shall be in accordance with the requirements of section 5 and the documents specified herein.

5. PREPARATION FOR DELIVERY

(The preparation for delivery requirements specified herein apply only for direct Government procurements. For the extent of applicablity of the preparation for delivery requirements of referenced documents listed in section 2, see 6.3.)

5.1 Preservation-packaging. Preservation-packaging shall be level A or C as specified (see 6.2).

5 1.1 Level A.

5.1.1.1 Cleaning. Fuseholders and associated fuse clips shall be cleaned in accordance with process C-1 of MIL-P-116.

5 1.1.2 Drying. Fuseholders and associated fuse clips shall be dried in accordance with MIL-P-116

5 1 1 3 Preservative application Preservatives are not used

5.1 1.4 Unit packaging Fuseholders and associated fuse clips shall be packaged in accordance with the methods specified herein of MIL-P-116, insuring compliance under the methods of preservation (unit protection) and the physical protection requirements paragraph therein

5 1.1 4 1 Fuseholders and associated fuse clips without silver plated parts.

5 1 1.4 1.1 Fuse clips. Fuse clips shall be packaged method III in quantities of five per unit package.

5 1.1.4 1.2 Fuseholders with no dimension exceeding 1 inch. Fuseholders shall be packaged method III in quantities of five per unit package.

5 1 1 4 1.3 Fuseholders exceeding 1 inch and up to and including 300 cubic inches or 5 pounds. Fuseholders shall be individually packaged method III

5.1 1 4.1.4 Fuseholders exceeding 300 cubic inches or 5 pounds. Fuseholders shall be individually packaged method III in a unit container conforming to PPP-B-636, class weather resistant, style optional, special requirement Containers shall be closed method V in accordance with the appendix to the container specification This container may serve as a shipping container

5 1 1 4 2 Fuseholders and associated fuse clips with silver plated parts

5 1 1.4 2 1 Fuse clips Unless otherwise specified (see 6.2), fuse clips shall be packaged method IA8 in quantities of five per unit package.

5.1 1 4 2.2 Fuseholders with no dimension exceeding 1 inch Unless otherwise specified (see 6 2), fuseholders shall be packaged method IA8 in quantities of five per unit package

5 1 1 4 2 3 Fuseholders exceeding 1 inch and up to and including 300 cubic inches or 5 pounds Unless otherwise specified (see 6 2), fuseholders shall be individually packaged method IA8 Unit packages which exceed 20 cubic inches shall be placed in a supplementary container conforming to PPP-B-566 or PPP-B-676

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5.1.1.4.2.4 Fuseholders exceeding 300 cubic inches or 5 pounds. Fuseholders shall be individually packaged method IA14. The exterior container shall conform to PPP-B-636, class weather resistant, style optional, special requirement. Containers shall be closed method V in accordance with the appendix to the container specification. This container may serve as a shipping container.

5.1.1.5 Intermediate packaging. Fuseholders and associated fuse clips, packaged as described herein, and whose outer container is not a box, shall be placed in intermediate containers conforming to PPP-B-566 or PPP-B-676. Intermediate containers shall be uniform in size and shape, shall be of minimum tare and cube, and shall contain multiples of five unit packages, not to exceed 100 packages or ten pounds. No intermediate packaging is required when the total quantity shipped to a single destination is less than 50 units.

5.1.2 Level C. Cleaned and dry fuseholders and associated fuse clips shall be packaged in quantities specified (see 6.2), in a manner that will afford adequate protection against corrosion, deterioration, and physical damage during shipment from supply source to the first receiving activity. The suppliers normal retail or wholesale preservation-packaging methods may be utilized when such meets the requirements of this level.

5.2 Packing. Packing shall be level A, B, or C, as specified (see 6.2).

5.2.1 Level A. The packaged fuseholders and associated fuse clips shall be packed in fiberboard containers conforming to PPP-B-636, class weather resistant, style optional, special requirement. Containers shall be closed method V in accordance with the appendix to the container specification.

5.2.2 Level B. The packaged fuseholders and associated fuse clips shall be packed in fiberboard containers conforming to PPP-B-636, class domestic, style optional, special requirement. Containers shall be closed method I in accordance with the appendix to the container specification. For Army, fiberboard containers shall be class weather resistant, as specified in level A.

5.2.3 Level C. The packaged fuseholders and associated fuse clips shall be packed in shipping containers in a manner that will afford adequate protection against damage during direct shipment from the supply source to the first receiving activity. Container packing or method of shipment shall comply with Uniform Freight or National Motor Freight Classification Rules or Regulations or other carrier rules as applicable to the mode of transportation.

5.3 Marking. In addition to any special marking required by the contract, each unit package, intermediate, and exterior container shall be marked in accordance with MIL-STD-129.

5.4 General. Exterior containers shall be of a minimum tare and cube consistent with the protection required and shall contain equal quantities of identical stock numbered items to the greatest extent possible.

6 NOTES

6.1 Intended use. Fuseholders and associated fuse clips covered by this specification are for use with electrical fuses primarily covered by MIL-F-15160 and MIL-F-23419, in overload and short circuit protection of power, and branch circuits associated with electrical, electronic, telephone and instrument circuits.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Title, number, and date of the applicable specification sheet, and the part number (see 1.2.1 or 1.2.2, and 3.1).
(c) When first article is required for inspection and approval (see 3.2, 4.5, and 6.7).
(d) Stud material when other than as specified (see 3.4.2).
(e) Submission of tested sample, if different from that specified (see 4.1.1.3).
(f) Levels of preservation-packaging and packing, and applicable marking (see section 5.1 and 5.2).
(g) Method of preservation-packaging, if other than method IA8 (see 5.1.1.4.2.1, 5.1.1.4.2.2, and 5.1.1.4.2.3).
(h) Special marking if required (see 5.3).

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6 2 1 Data requirements When this specification is used in a procurement which incorporates a DD Form 1423 and invokes the provisions of 7-104 9(n) of the Armed Services Procurement Regulations, the data requirements identified below will be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved Contract Data Requirements List (DD Form 1423) incorporated into the contract When the provisions of ASPR-7-104.9(n) are not invoked, the data specified below will be delivered by the contractor in accordance with the contract requirements Deliverable data required by this specification is cited in the following paragraphs

Paragraph	Data Requirement	Applicable DID
4.1.1.3	First article Inspection	UDI-T-23450
4 3	Report, Material Inspection System	UDI-T-23741

(Copies of data item descriptions required by the contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

6.3 Sub-contracted material and parts. The preparation for delivery requirements of referenced documents listed in Section 2 do not apply when material and parts are procured by the contractor for incorporation into the equipment and lose their separate identity when the equipment is shipped.

6 4 Electrical rating. The voltage rating specified (see 3 1) is the maximum d.c or a.c root mean square (rms) voltage for which a fuseholder is designed. The current rating specified (see 3.1) is the maximum current for which a fuseholder of fuse clip is designed

6.5 Grouping of metals (see 3.3 3 1). The grouping in table VIII is intended to serve as a guide in selecting materials to be used in electronic equipment, and should not be construed to waive requirements herein or in the applicable specification sheet as to corrosion resistance of parts and assemblies

TABLE VIII - Grouping of metals.

Group I	Group II	Group III	Group IV
Magnesium alloys (most anodic)	Aluminum Aluminum alloys Zinc Cadmium Tin Corrosion-resistant steel	Zinc Cadmium Steel Lead Tin Corrosion-resistant steel	Copper and its alloys Nickel and its alloys Chromium Corrosion-resistant steel Gold Silver (most cathodic)

6 5 1 Metals application Unless specifically approved by the Government, all metals not listed in table VIII should be considered dissimilar with respect to one another and with respect to any of the materials listed. Except for zinc, cadmium, and tin as listed in groups II and III, and for corrosion-resistant steel as listed in groups II, III, and IV, contact between a member of one group and a member of any other group should be considered dissimilar Such contact should not be made unless necessary, in which case it should be demonstrated that the contact is not detrimental, or an interposing material should be used which will be compatible to each metal. When reference is made to a metal in a particular group, the reference applies to the metal on the surface of the part that is zinc means zinc coating as well as zinc electroplate, zinc hot-dip, or zinc metal-spray Different metals in contact, even though similar, should be employed in assemblies in such a manner that the smaller part is cathodic (or protected) and the larger part is anodic (or corroded) if any corrosion takes place Care should be exercised in using different aluminum alloys against each other or against differing material

6.6 Warning Potentially hazardous situations could develop in some of the test procedures specified in this specification Precautions should therefore be taken to insure that test personnel are adequately protected and observe the necessary safety measures at all times

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6.7 First article. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection as to those bidders offering a product which has been previously procured or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending procurement.

6.8 Changes from previous issue. "#" are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Custodians
Army - EL
Navy - SH
Air Force - 85

Review activities
Army - MI, MU, EA
Navy - EC
Air Force - 80, 17
DSA - ES

User activities
Army - WC, AT.
Navy - MC, AS, CG
Air Force - 19

Preparing activity
Navy - SH
(Project 5920-0319)

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CENTER BUILDING - SEC 6124
PRINCE GEORGES CENTER
HYATTSVILLE, MARYLAND 20782

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

POSTAGE AND FEES PAID
DEPARTMENT OF NAVY

DOD 316



COMMANDER
NAVAL SHIP ENGINEERING CENTER
CENTER BUILDING - SEC 6124
PRINCE GEORGES CENTER
HYATTSVILLE, MARYLAND 20782

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OMB Approval
No 22-R255

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DOCUMENT IDENTIFIER AND TITLE

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1 HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A GIVE PARAGRAPH NUMBER AND WORDING

B RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2 COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3 IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO *(If "Yes" in what way?)*

4 REMARKS

SUBMITTED BY *Printed or typed name and address - Optional*

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DATE

DD FORM 1426 JAN 72

REPLACES EDITION OF 1 JAN 66 WHICH MAY BE USED

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